Wine Quality Prediction Using a Weighted Ensemble

Reproduction, class balancing, feature importance and a web application

Base paper: Ma et al. (2024), From vineyard to table — Journal of Business Research 176:114576

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Guide
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Final Review

The question: can lab tests predict what tasters will say?

The business problem behind the base paper

11 lab measurements

Alcohol, acidity, sulphates, sugar, sulphur dioxide, pH, density …

Cheap, objective, available early

A trained model

Learns the link between the chemistry and the grade from past wines

This project

Quality grade 3 – 8

Normally given by a panel of tasters

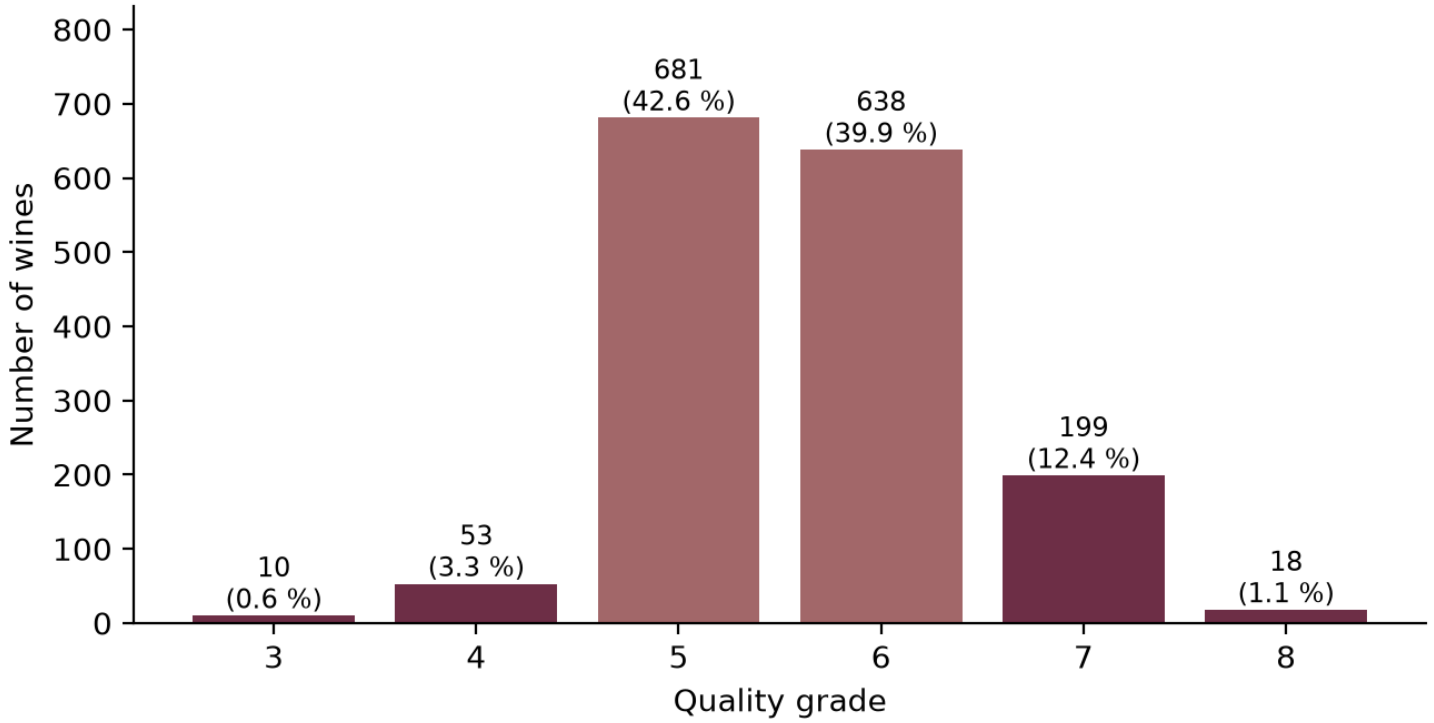
Slow, costly, partly subjective

Why a wine retailer cares

- Decide prices, stock and promotions before the tasting panel has met
- Check every batch the same way, every time
- Know which measurements actually drive quality — and which tests can be skipped

The dataset: UCI Wine Quality (red)

Red Vinho Verde wines from Portugal — collected by Cortez et al. (2009)



1,599
WINES (ROWS)

11
LAB MEASUREMENTS

6
GRADES (3 TO 8)

82.5%
ARE GRADE 5 OR 6

The difficulty is the imbalance. Grade 3 has only 10 wines and grade 8 only 18 (1.75% together). A model that always answers "5" is already right 42.6% of the time — while never recognising a single very poor or excellent wine, which are exactly the ones a retailer most wants to find.

The base paper

Ma, Mao, Cao, Luo, Gupta & Wang (2024), Journal of Business Research 176:114576 (Elsevier)

SVM

RNN

ELM

Random Forest

Two weights per model

Accuracy weight — the more accurate the model, the bigger its say.

Diversity weight — the more its mistakes differ from the others', the bigger its say.

Final weight = average of the two.

Weighted vote

→ one grade

89.73%

ACCURACY CLAIMED BY THE PAPER

+9.02

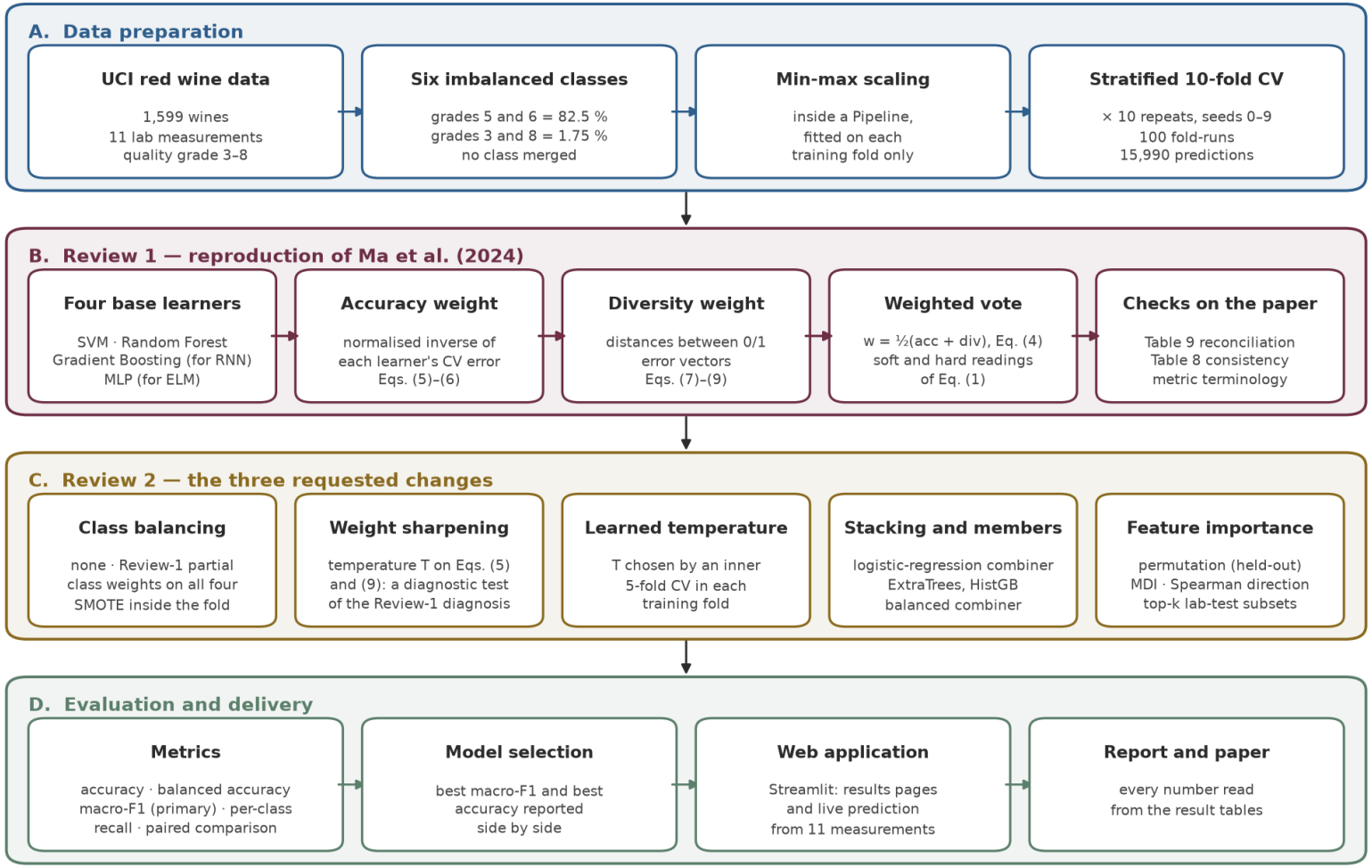
POINTS ABOVE ITS BEST SINGLE MODEL

The paper also reports that alcohol, sulphates, volatile acidity and citric acid matter most.

An ensemble is a team of models that vote. The paper's claim is that this team is far better than its best member (89.73% vs 80.71%). Typical published accuracies for this six-grade task are in the 60s to low 70s — so the first job was to check whether the claim holds.

The whole project on one page

Four parts, three milestones



1

Review 1

Rebuild the paper's method exactly, test it fairly, and check the paper's own numbers

2

Review 2

The three changes the guide asked for: class balancing, a better result, feature importance

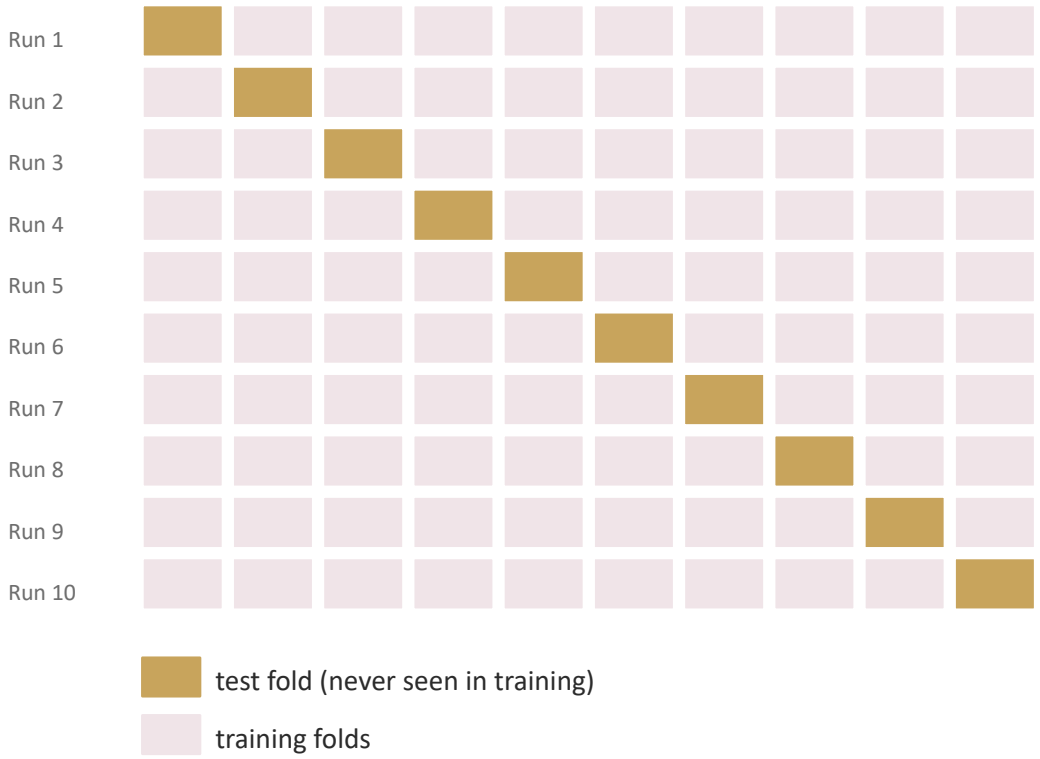
3

Final

A web application, the project report, a conference paper, and the code on GitHub

How every model is tested — fairly

Stratified 10-fold cross-validation, repeated 10 times



… then reshuffle and repeat the whole thing 10 times = 100 fold-runs

- The wines are split into 10 groups with the same grade mix (stratified); each group takes a turn as the test set
- Every wine is predicted 10 times by a model that never saw it: 10 × 1,599 = 15,990 predictions per model
- Every model uses exactly the same splits, so models can be compared repeat by repeat
- No data leakage: scaling, SMOTE and feature selection are all learned inside the training folds only

Leakage = peeking at the exam answers

An early version scaled all the data before splitting it — the test wines quietly shaped the training. It was fixed at Review 1 by putting the scaler inside a pipeline.

The four base learners

Two kept from the paper, two replaced to stay within the classical-ML syllabus

Paper's model	Our model	The idea in one line	Accuracy (ours)
SVM	Support Vector Machine (RBF kernel)	Draws the widest possible boundary between the grades	61.95%
Random Forest	Random Forest (300 trees)	Many decision trees, each trained on a random sample, vote together	71.03%
RNN (recurrent neural network)	Gradient Boosting (200 stages)	Each new tree corrects the mistakes of the trees before it	67.67%
ELM (extreme learning machine)	Multilayer Perceptron (64, 32)	A small neural network trained by back-propagation	61.35%

Why two substitutes?

RNNs and ELMs are outside the classical machine-learning syllabus of the course. Gradient Boosting also improves step by step, like the RNN; the MLP is the closest classical neural network to the ELM.

Same set-up for all four

Each model is a pipeline: min–max scaling, then the model. SVM and Random Forest use balanced class weights so the rare grades are not ignored. Every model gives class probabilities, which the weighted vote needs.

The paper's weighted vote — and what the weights turn out to be

Implemented exactly as the paper's equations describe

1 **Accuracy weight**

1 ÷ the model's error rate, normalised so the four add up to 1 (paper's Eqs. 5–6)

2 **Diversity weight**

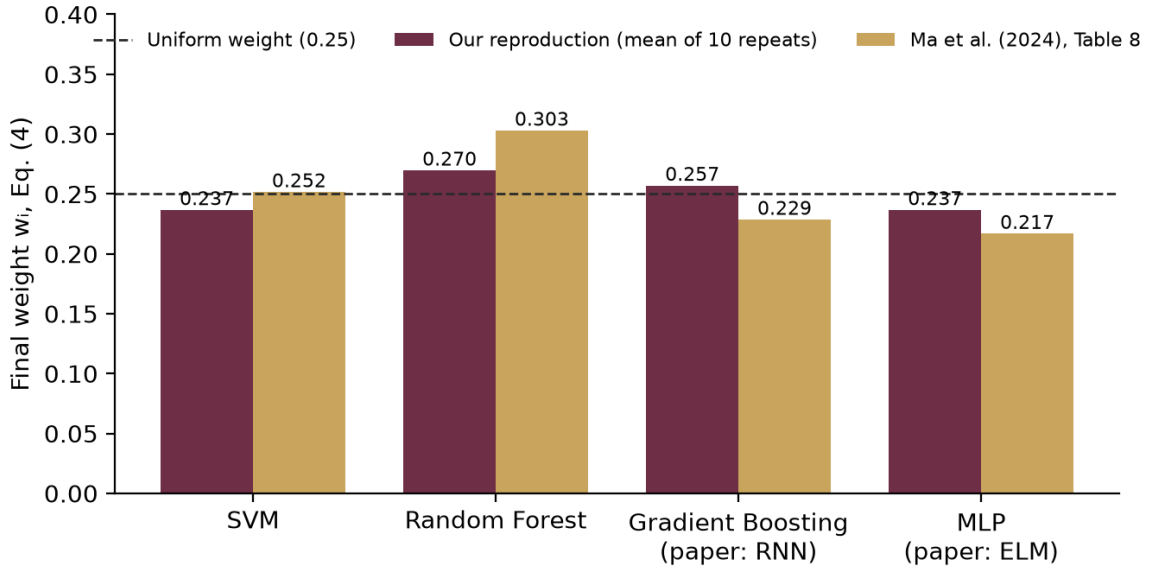
How far the model's list of mistakes is from the others' lists, normalised (Eqs. 7–9)

3 **Final weight**

The average of the two (Eq. 4)

4 **Weighted vote**

Add up each model's probabilities × its weight; the highest grade wins (Eq. 1)

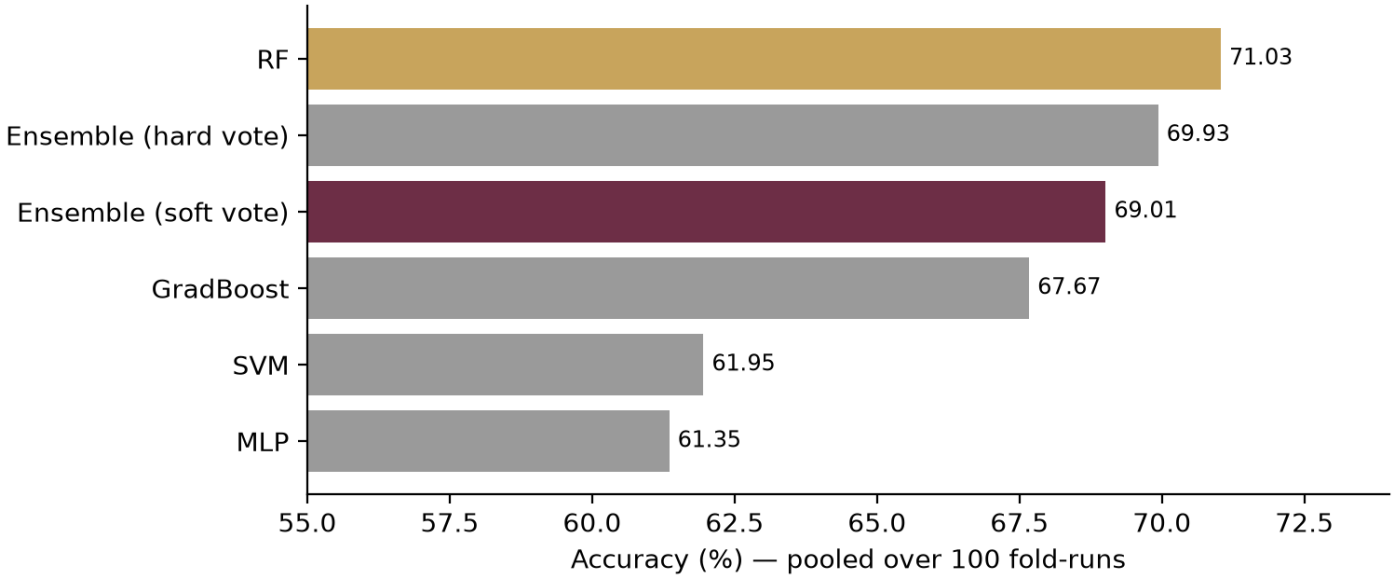


Our weights: 0.237 – 0.270. The paper's own: 0.217 – 0.303. Both stay close to 0.25 = an equal share. With equal weights a vote is just an average — and an average of one strong and three weaker models lands between them, not above the best.

Review 1 — the reproduction result

Pooled over 100 fold-runs (15,990 predictions per model)

89.73%	69.01%	71.03%	0.0% / 0.6%
PAPER'S ENSEMBLE (CLAIMED)	OUR ENSEMBLE (FAITHFUL REPRODUCTION)	RANDOM FOREST ALONE — BEATS THE ENSEMBLE	GRADE-3 / GRADE-4 WINES FOUND



- The paper's central claim does not reproduce: our ensemble is 2.02 points below Random Forest, not 9 points above
- It does beat the other three models — exactly what an average does
- Grades 5 and 6 are predicted well; the rare grades are almost never found

Why 89.73% does not reproduce

Three checks computed from the paper's own published tables

1 Different test set

The paper's confusion matrix (Table 9) holds only 491 wines — 30.7% of the 1,599. That is one hold-out split, not 100 cross-validation runs.

2 Its own arithmetic

The correct predictions in that table: 403 of 491 = 82.08% — 7.65 points below the 89.73% in the text.

3 Flat weights

The paper's weights (Table 8) applied to its own model accuracies give 67.52%. Our 69.01% sits 1.49 points from that; the paper's claim sits 22.21 points from it.

Our explanation: the weighting gives every model about a quarter of the vote, so the ensemble averages its members. Our result lands where that predicts — and inside the range other studies report for this data.

Caveat: the Table 9 cells were read by eye from the printed PDF. These are figures we could not reconcile — not a claim that the paper is wrong.

Review 2 — the three changes the guide asked for

Same data, same folds, same rules as Review 1, so every number is comparable

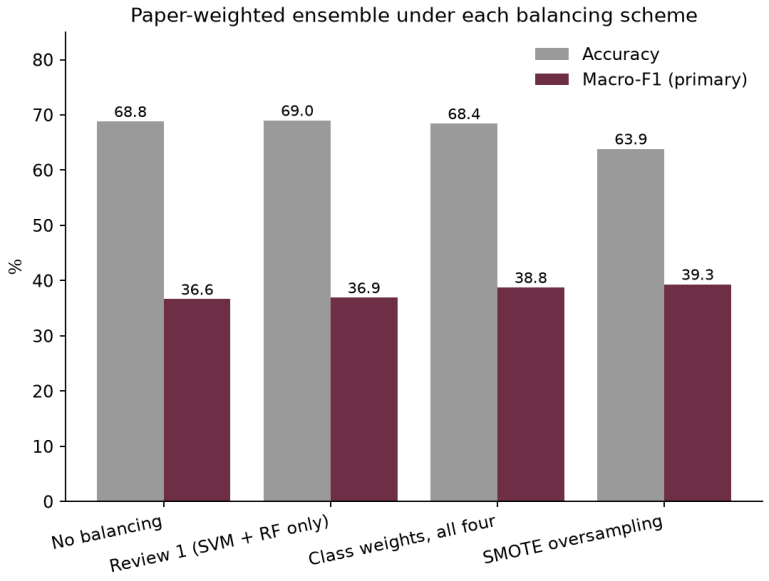
Asked at Review 1	What was done	Result in one line
1. Class balancing — yet to be done	Four schemes compared: none, Review-1 partial, class weights on all four, SMOTE inside the fold	Macro-F1 36.92 → 39.34 (paper ensemble, SMOTE)
2. Improve the result	Tested the flat-weights explanation, then fixed it by learning the weight sharpness inside each training fold; also stacking and stronger models	Paper's ensemble 69.01% → 70.89%; best macro-F1 40.92
3. Work on feature importance	Permutation importance on held-out folds, checked against two other methods and the paper	alcohol, sulphates, volatile acidity lead; 4 tests ≈ all 11

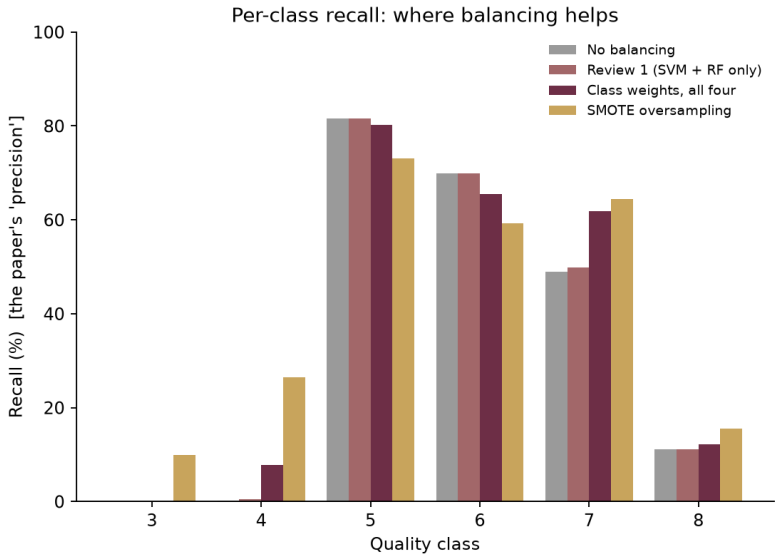
Judged on macro-F1 — fixed in the code before any experiment ran

Accuracy mostly measures grades 5 and 6. Macro-F1 scores each of the six grades separately and averages them with equal weight, so a model only does well if it also finds the rare grades. Accuracy is still reported everywhere, for comparison with the paper.

Change 1 — class balancing

The paper's weighted vote under four schemes — only the balancing changes





What it shows

- Class weights make mistakes on rare grades cost more; SMOTE creates new rare-grade examples by blending two similar real wines
- SMOTE gives the best macro-F1: grade 4 found 0.6% → 26.4%, grade 7 49.9% → 64.5%
- The price: accuracy −5.15 points, because some grade-5/6 wines are now called rare
- SMOTE runs inside each training fold — test wines are never copied into training

Scheme	Accuracy	Macro-F1	Grade 3	Grade 4	Grade 7	Grade 8
No balancing	68.82	36.64	0.0	0.2	49.0	11.1
Review 1 (SVM + RF only)	69.01	36.92	0.0	0.6	49.9	11.1
Class weights, all four	68.43	38.77	0.0	7.7	61.9	12.2
SMOTE oversampling	**63.86**	**39.34**	**10.0**	**26.4**	**64.5**	**15.6**

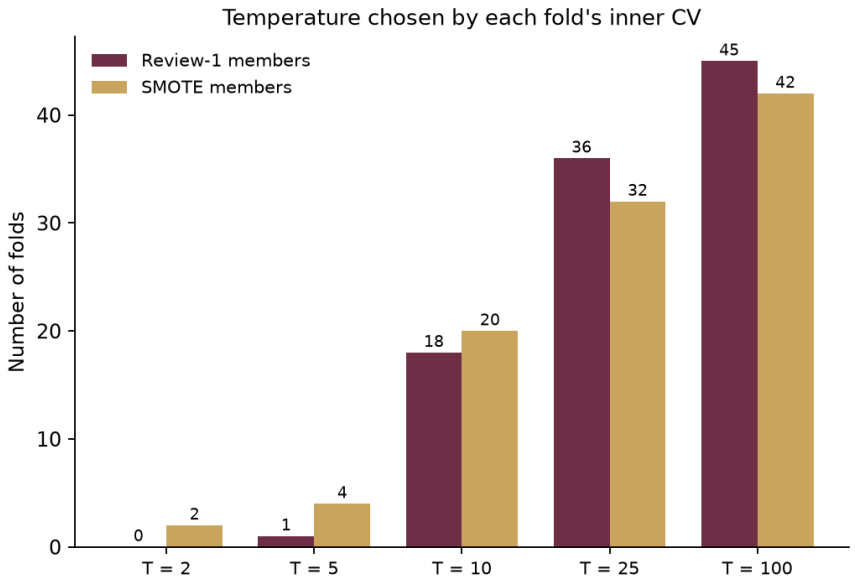
Grade columns = share of that grade's wines found (recall, %)

Change 2 — improving the result

Test the flat-weights explanation, then fix it without touching the test data

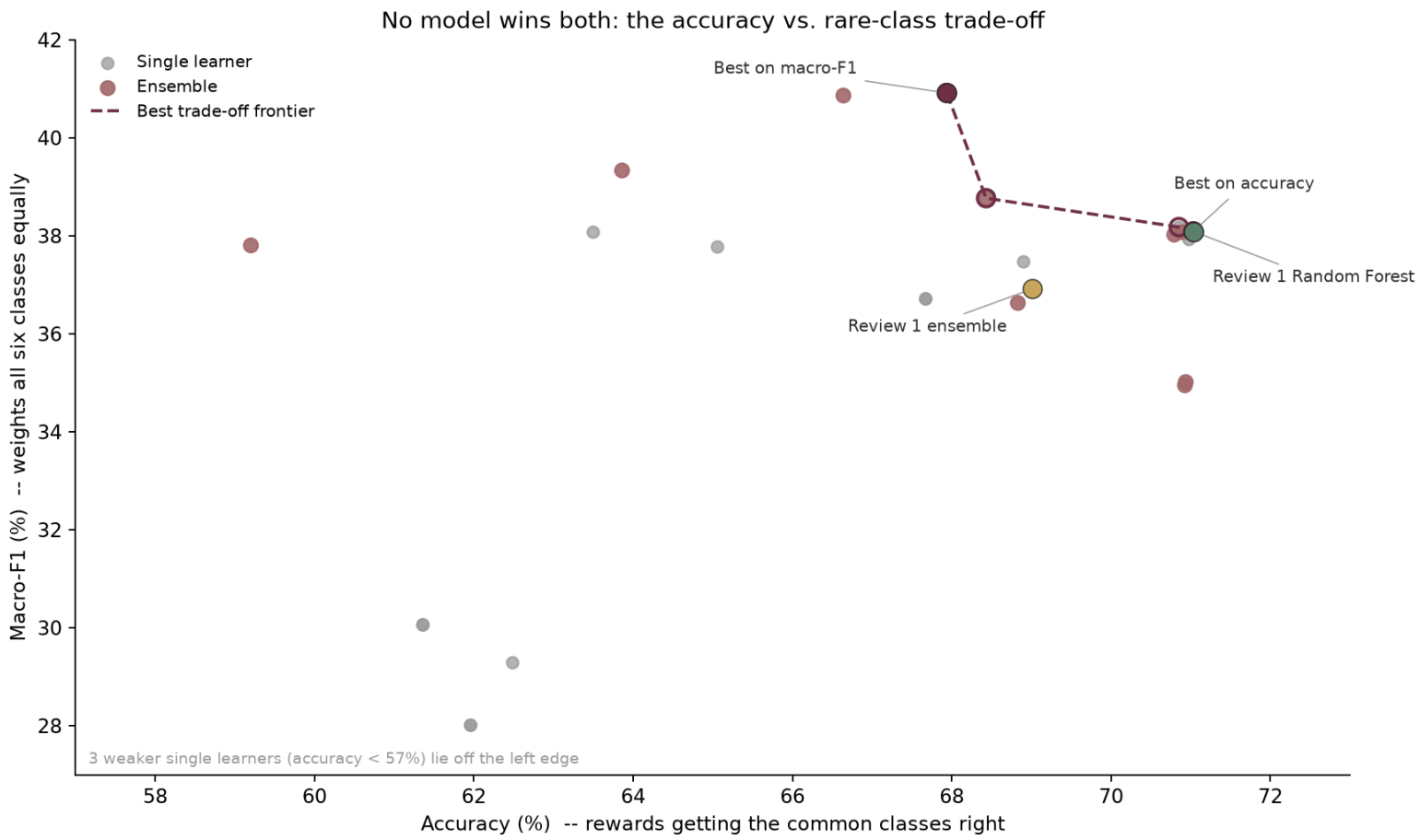
69.01%	70.89%	71.03%	99%
PAPER'S WEIGHTS (T = 1)	LEARNED SHARPNESS, SAME FOUR MODELS	RANDOM FOREST ALONE	OF FOLDS CHOSE A SHARP T ≥ 10

- Idea: raise the weights to a power T before normalising — T = 1 is the paper, a bigger T gives more say to the best model
- Test: with a big enough T (≥ 25) the ensemble overtakes Random Forest — the explanation holds (a diagnostic only)
- Fix: choose T inside each training fold by an inner 5-fold CV. Better than the paper's weights in 10 of 10 repeats; now level with Random Forest
- Also tried: stacking (70.92% accuracy but macro-F1 only 34.96 — it relearns the majority bias) and stronger models (ExtraTrees, HistGB)



All 28 candidates — no model wins both measures

Each point is one model, scored on the same 100 fold-runs



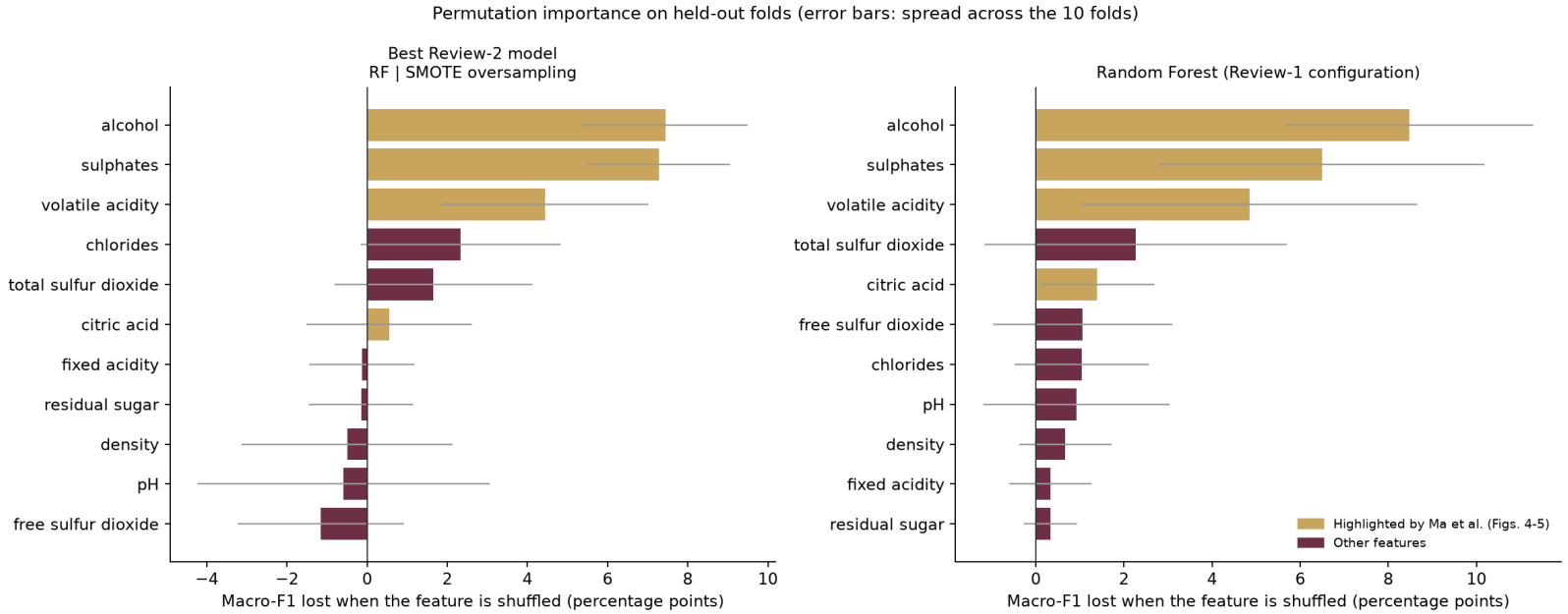
Reading the chart

- Right = better accuracy (mostly grades 5 and 6); up = better on all six grades equally
- Dashed line: models that no other model beats on both at once
- Best macro-F1: Random Forest · SMOTE — 40.92 (accuracy 67.94%)
- Best accuracy: Random Forest · Review-1 setup — 71.03%
- Random Forest + SMOTE beats the Review-1 ensemble on macro-F1 in 10 of 10 repeats

So the web application offers both: the most accurate model, and the one that finds rare grades best.

Change 3 — which measurements drive the prediction?

Permutation importance: shuffle one measurement in the test fold and see how much the score drops



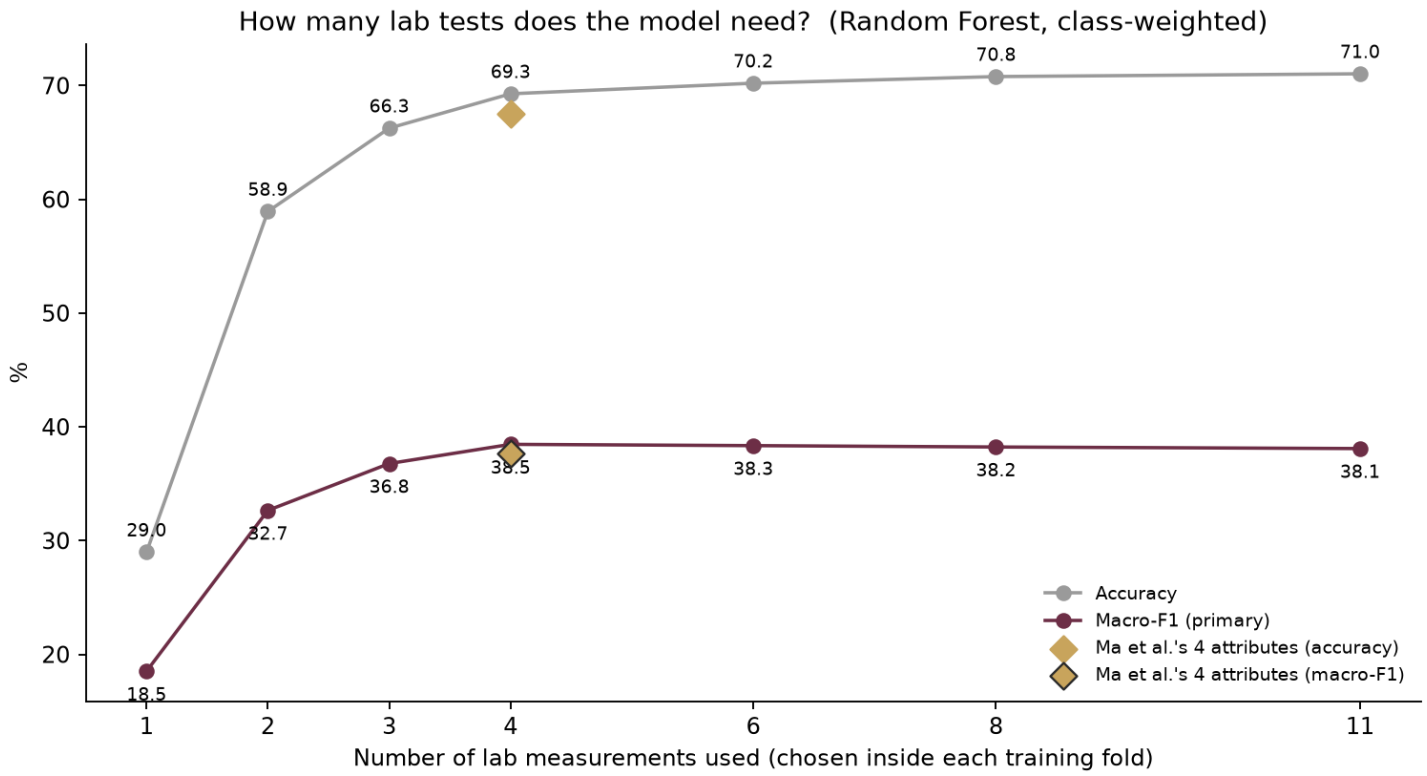
What it shows

- Top three: alcohol, sulphates, volatile acidity
- Three methods agree (rank agreement 0.74 – 0.89)
- Better wine: more alcohol, sulphates, citric acid. Worse wine: more volatile acidity
- 3 of the paper's 4 key attributes confirmed; citric acid ranks #6

Method	What it measures	Role
Permutation importance	Score lost when one measurement is scrambled on unseen wines	Main measure
Random Forest impurity (MDI)	How much the forest's splits use each measurement	Cross-check (known bias)
Spearman correlation	Whether more of it means a better or a worse wine	Direction

How many lab tests does the model really need?

Random Forest retrained on the top-k measurements, chosen inside each training fold



Business reading

- 4 tests (volatile acidity; sulphates; alcohol; chlorides) give 69.27% accuracy — 98% of what all 11 give (71.03%)
- With 4 tests macro-F1 is 38.46, slightly above all 11 (38.09)
- The paper's own four attributes give 67.55%
- A winery could screen batches with a much smaller set of tests

Review 1 → Review 2: before and after

Best model on the main measure: Random Forest · SMOTE

36.92 → 40.92

MACRO-F1 (MAIN MEASURE)

35.52 → 42.07

BALANCED ACCURACY

69.01 → 67.94

ACCURACY (THE PRICE)

	Review-1 ensemble	Review-1 Random Forest	Review-2 best	Change
Grade 3 found (%)	0.00	0.00	4.00	+4.00
Grade 4 found (%)	0.57	0.00	21.89	+21.32
Grade 5 found (%)	81.66	81.97	78.00	−3.66
Grade 6 found (%)	69.87	74.29	64.19	−5.69
Grade 7 found (%)	49.90	51.06	65.43	+15.53
Grade 8 found (%)	11.11	11.11	18.89	+7.78

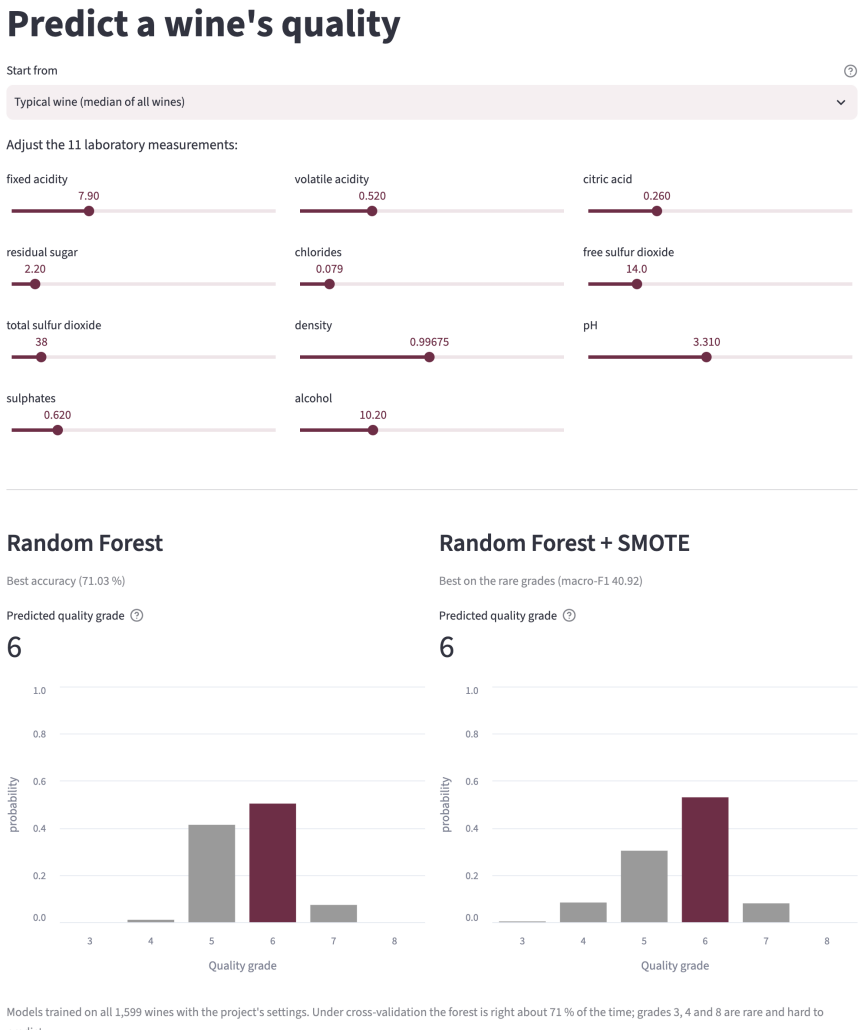
In one sentence

Balancing moves accuracy away from the two crowded grades towards the rare ones: more rare wines are found, for a small loss overall. Nothing reaches the paper's 89.73% — the best accuracy of any fair configuration is 71.03%.

The web application

Built with Streamlit on the project's own code — so it always shows the project's numbers

- Seven pages: overview, dataset, Review 1, class balancing, improving the result, feature importance, prediction
- Predict a wine: set 11 sliders (or start from a typical wine of any grade) and get the grade with the probability of every grade
- Two models side by side: Random Forest (most accurate, 71.03%) and Random Forest + SMOTE (best on rare grades)
- Batch mode: upload a CSV of wines, download the predictions
- 15 automated checks: every page loads, the models match the project's exactly, the numbers match the result files — all pass



Run it

```
streamlit run app/streamlit_app.py
Code: github.com/Surianandhan/wine-quality-ml-project
```

What the project delivers

Everything is generated from the same result files, so it cannot disagree

Code and results

Python modules, 100-fold-run experiments, every table and figure

github.com/Surianandhan/...

Web application

Seven pages, live prediction, batch CSV, 15 automated checks

Streamlit

Project report

69 pages in the SASTRA format: 9 chapters, 33 verified references

Word + PDF

Conference paper

10 pages: method, results, discussion

Word + PDF

Colab notebook

Runs the whole project in a browser, no installation

Google Colab

One command

python run_all.py --review2
reruns every experiment and rewrites every table and figure.

Conclusion

Reproduction

The paper's method, rebuilt exactly and tested fairly, reaches 69.01% — not 89.73%. Random Forest alone does better (71.03%).

Why

The paper's weights are nearly equal, so the ensemble averages its members; the paper's own tables point the same way.

Three changes

Balancing finds far more rare wines (macro-F1 36.92 → 40.92); learned weights lift the paper's ensemble to 70.89%; alcohol, sulphates and volatile acidity drive quality.

Delivered

A tested web application, a 69-page report, a conference paper and reproducible code.

Lesson: an ensemble only beats its best member if the weights let the best member lead.

Limitations and future work

Stated openly

Limitations

- Two of the paper's four models had to be replaced by syllabus-friendly substitutes
- Grade 3 has only 10 wines and grade 8 only 18 — too few for any model to learn reliably
- The best model is picked on the same cross-validation that scores it — a small optimistic bias
- The 10 repeats reuse the same wines, so the statistical test across them is indicative only
- Only red wine, and only lab measurements — no price, region or vintage

Future work

- Repeat on the 4,898 white wines of the same dataset
- Treat the grade as ordered: predicting 6 for a 7 should cost less than predicting 3
- Nested cross-validation around the model choice, with proper confidence intervals
- Explain each individual prediction inside the web app

References

[1] Ma, R., Mao, D., Cao, D., Luo, S., Gupta, S., & Wang, Y. (2024). From vineyard to table: Uncovering wine quality for sales management through machine learning. Journal of Business Research, 176, 114576.

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[8] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. Journal of Machine Learning Research, 12, 2825–2830.

[9] Cortez, P., et al. (2009). Wine Quality [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C56S3T

Thank you — questions are welcome.

Backup 1 — the measures, in plain words

For one grade at a time, e.g. grade 4

Measure	Plain meaning	Formula
Accuracy	Share of all wines given the right grade	correct ÷ all
Recall (the paper's "precision")	Of the wines that really are grade 4, how many did we find?	TP ÷ (TP + FN)
Precision	Of the wines we called grade 4, how many really are?	TP ÷ (TP + FP)
F1	Balance of precision and recall for one grade	2PR ÷ (P + R)
Macro-F1 (main measure)	F1 of each of the six grades, averaged with equal weight	mean of 6 F1s
Balanced accuracy	Recall of each grade, averaged with equal weight	mean of 6 recalls

TP = right calls of grade 4 · FN = grade-4 wines we missed · FP = other wines wrongly called grade 4. The paper's per-class "precision" divides by the true grade totals — that is recall; the report notes this.

Backup 2 — top candidates, ranked by macro-F1

Pooled over 100 fold-runs

#	Model	Kind	Accuracy	Bal. acc.	Macro-F1
1	Random Forest · SMOTE	single learner	67.94	42.07	40.92
2	Learned-T ensemble · SMOTE	ensemble	66.63	42.49	40.87
3	Paper ensemble · SMOTE	ensemble	63.86	41.44	39.34
4	Paper ensemble · class weights	ensemble	68.43	37.94	38.77
5	Random Forest · no balancing	single learner	70.85	36.70	38.18
6	Random Forest · Review-1 setup	single learner	71.03	36.40	38.09
7	Random Forest · class weights	single learner	71.03	36.40	38.09
8	Gradient Boosting · SMOTE	single learner	63.49	39.40	38.08
9	Learned-T ensemble · Review-1 setup	ensemble	70.89	36.58	38.08
10	Paper weights: SVM + RF + ExtraTrees + HistGB	ensemble	70.79	36.96	38.04
11	ExtraTrees	single learner	70.97	36.45	37.93
12	Stacked: paper's 4 learners, balanced combiner	ensemble	59.20	50.03	37.82
13	Gradient Boosting · class weights	single learner	65.05	37.92	37.78
14	HistGB	single learner	68.89	36.86	37.48

Top 14 of 28. Full table: results/review2/tables/model_comparison.csv

Backup 3 — is the improvement consistent?

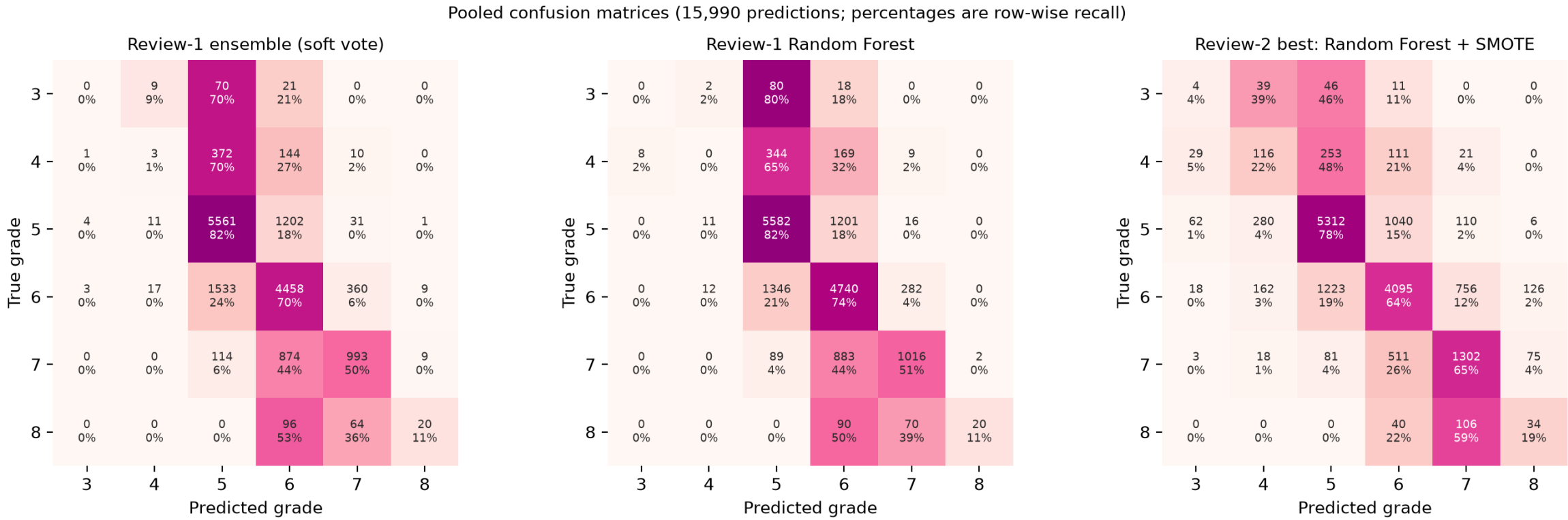
Paired comparison across the 10 repeats — every model scored on identical folds

Model	Compared with	Measure	Mean diff.	Repeats won	Wilcoxon p
Random Forest · SMOTE	Paper ensemble · Review-1 setup	Macro-F1	+3.97	10 / 10	0.002
Random Forest · SMOTE	Paper ensemble · Review-1 setup	Accuracy	−1.08	1 / 10	0.004
Random Forest · SMOTE	Random Forest · Review-1 setup	Macro-F1	+2.80	10 / 10	0.002
Random Forest · SMOTE	Random Forest · Review-1 setup	Accuracy	−3.10	0 / 10	0.002
Random Forest · SMOTE	Learned-T ensemble · SMOTE	Macro-F1	+0.01	4 / 10	0.846
Random Forest · SMOTE	Learned-T ensemble · SMOTE	Accuracy	+1.31	9 / 10	0.006
Learned-T ensemble · Review-1 setup	Paper ensemble · Review-1 setup	Macro-F1	+1.16	10 / 10	0.002
Learned-T ensemble · Review-1 setup	Paper ensemble · Review-1 setup	Accuracy	+1.88	10 / 10	0.002
Learned-T ensemble · Review-1 setup	Random Forest · Review-1 setup	Macro-F1	−0.01	6 / 10	0.922
Learned-T ensemble · Review-1 setup	Random Forest · Review-1 setup	Accuracy	−0.14	4 / 10	0.438
Random Forest · Review-1 setup	Paper ensemble · Review-1 setup	Macro-F1	+1.17	10 / 10	0.002
Random Forest · Review-1 setup	Paper ensemble · Review-1 setup	Accuracy	+2.02	10 / 10	0.002

The repeats reuse the same 1,599 wines with different splits, so they are not independent and the p-values are optimistic. The win count is the plainer evidence.

Backup 4 — confusion matrices

Rows = true grade, columns = predicted grade; percentages are the share of each true grade



Most mistakes go to a neighbouring grade. Balancing moves predictions out of the crowded 5 and 6 columns into the rare grades: more rare wines caught, and some common wines now called rare.

Backup 5 — feature-importance table

Macro-F1 points lost when the measurement is shuffled; Spearman ρ with quality

Rank	Measurement	Best model	Random Forest	RF impurity	ρ with quality	Paper key?
1	alcohol	7.45	8.48	0.135	0.48	yes
2	sulphates	7.28	6.49	0.119	0.38	yes
3	volatile acidity	4.44	4.86	0.129	-0.38	yes
4	chlorides	2.33	1.05	0.093	-0.19	
5	total sulfur dioxide	1.66	2.27	0.089	-0.20	
6	citric acid	0.56	1.39	0.079	0.21	yes
7	fixed acidity	-0.13	0.34	0.068	0.11	
8	residual sugar	-0.15	0.34	0.065	0.03	
9	density	-0.50	0.67	0.085	-0.18	
10	pH	-0.59	0.92	0.076	-0.04	
11	free sulfur dioxide	-1.16	1.06	0.062	-0.06	